EXPERIMENT - 16

AIM :

To write and implement a Python program to create a Feed Forward Neural Network (FFNN) and demonstrate its working using sample input data.

ALGORITHM:

1. Import the required libraries such as NumPy and TensorFlow.
2. Create the input dataset and corresponding output values.
3. Initialize the Feed Forward Neural Network.
4. Add input, hidden, and output layers.
5. Compile the model using an optimizer and loss function.
6. Train the neural network using the training dataset.
7. Predict the output for the given input values.
8. Display the predicted results.

PROGRAM :

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```
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

# Input data
X = np.array([[0, 0],
              [0, 1],
              [1, 0],
              [1, 1]])

# Output data (XOR)
y = np.array([[0],
              [1],
              [1],
              [0]])

# Create Feed Forward Neural Network
model = Sequential()

# Input layer and hidden layer
model.add(Dense(4, input_dim=2, activation='relu'))

# Output layer
model.add(Dense(1, activation='sigmoid'))

# Compile the model
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])

# Train the model
model.fit(X, y, epochs=1000, verbose=0)

# Predict the output
predictions = model.predict(X, verbose=0)

# Display results
print("Predicted Output:")
for i in range(len(X)):
    print(X[i], ":", round(float(predictions[i][0])))
```

OUTPUT:



RESULT:

Thus, the Feed Forward Neural Network was successfully implemented using Python and TensorFlow. The network was trained using the XOR dataset and correctly predicted the output values